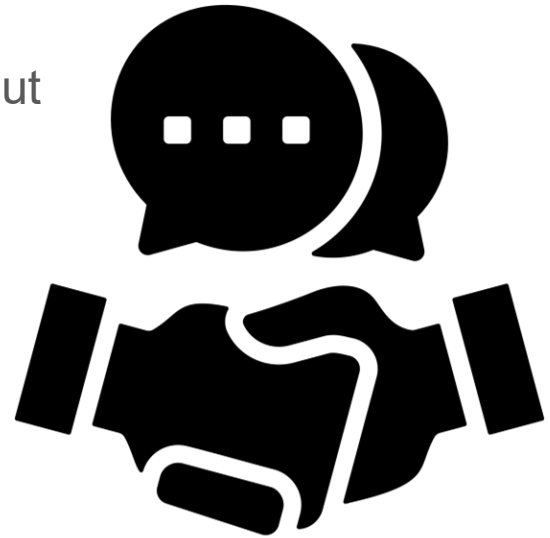
Unit 1:
Module Overview and
Introduction to Informatics for FETP

Introductions

What is your name?

What is one question or challenge you have about using technology in your work as a field epidemiologist?

What are your expectations for this module?



Why Are We Here?

Through the FETP Program, countries have increased capacity to prevent, detect, and respond to public health priority issues.

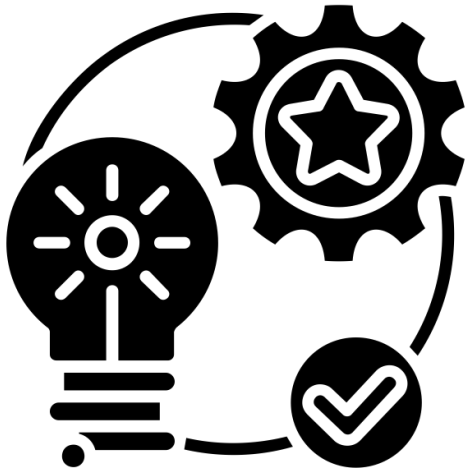
Field epidemiologists' ability to collect, analyze, interpret, and use data is critical to their everyday work.

Data is becoming increasingly available as technology evolves and transforms country health information systems.

Epidemiologists are faced with the need to rapidly navigate and use an array of evolving digital health tools

Informatics for FETP Competency

Field epidemiologists will be capable of **applying informatics principles, systems, and tools to public health surveillance, outbreak investigations, and public health action.**



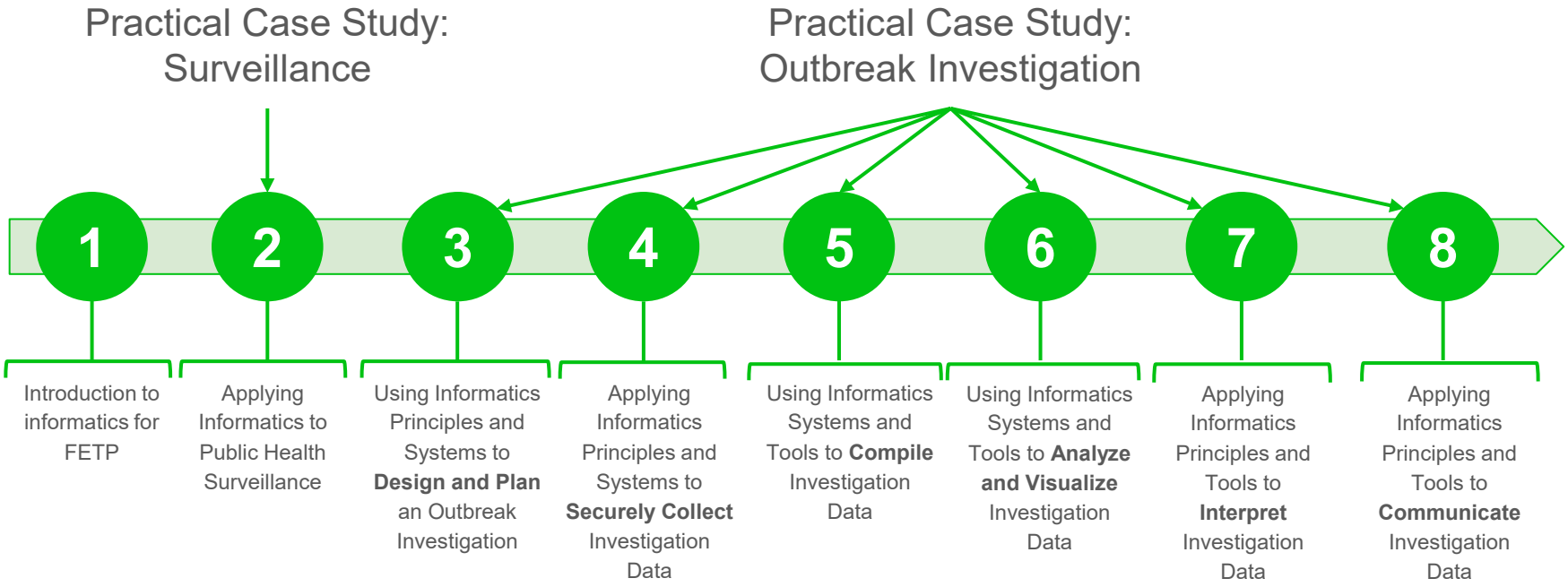
Informatics for FETP Objectives

1. Relate key informatics principles and systems to the public health surveillance cycle and system
2. Re-use data from existing information systems to detect, diagnose, and respond to outbreaks
3. Use common software features and functions to explore, describe, and prepare data for analysis
4. Identify and use software features and functions to detect and address data quality issues
5. Apply informatics standards and software features to organize and analyze surveillance data
6. Develop a plan for ongoing and systematic data collection, management, analysis, interpretation and dissemination that uses appropriate informatics principles, systems. and tools based on the outbreak investigation.

Informatics for FETP Objectives

7. Use informatics principles, systems, and standards to inform the selection, design, and use of appropriate data collection applications based on the specific outbreak and available resources
8. Apply informatics principles, systems, and tools to compile data from multiple sources and synthesize information to inform actions
9. Apply analytics and visualization to the discovery, interpretation, and use of public health data to extract meaningful insights from complex data to tackle a wide variety of public health challenges.
10. Use informatics tools to improve interpretation of results and inform outbreak control measures.
11. Use informatics principles and tools to communicate and disseminate clear and concise health information.

Informatics for FETP Program & Schedule

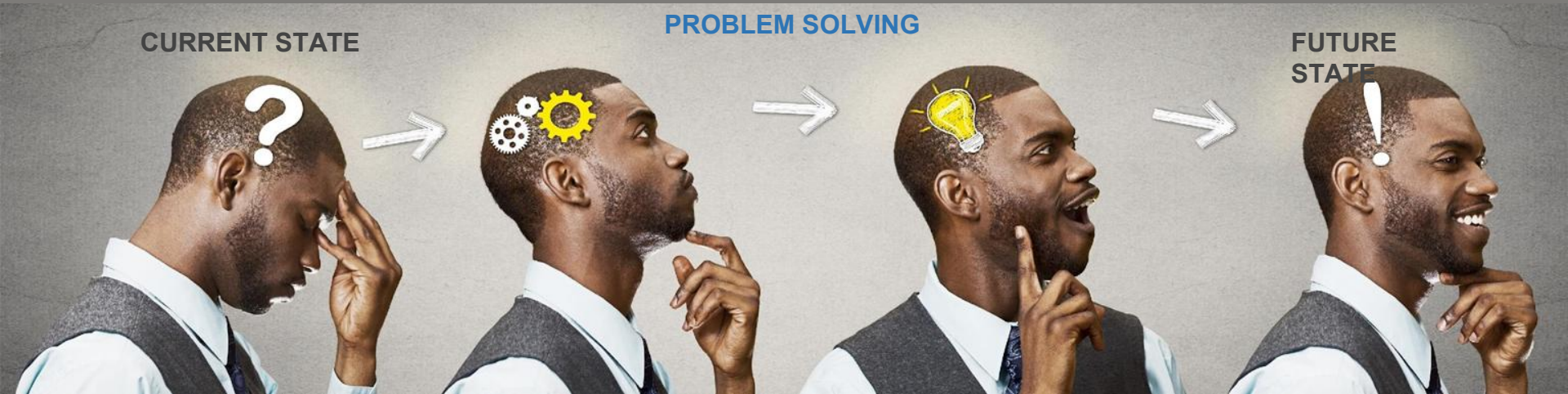


Problem-Solving Approach

Carrying out public health mission requires systematic identification of informatics problems in public health - and the development of the means to solve those problems.

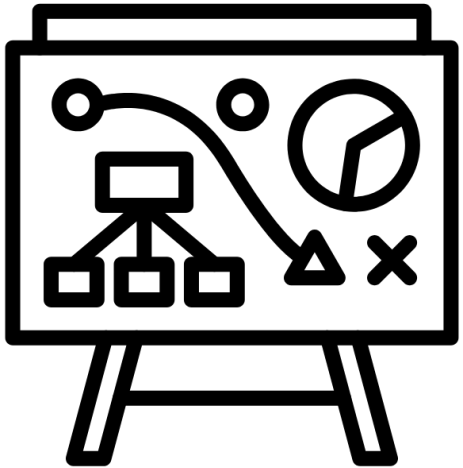
Each *Informatics for FETP* unit centers around specific problems that epis face in the real world - and that can be solved using informatics principles, systems, and tools.

Problem solving is a step-by-step, logical approach that takes us from our current state to a future one.



Learning Methods

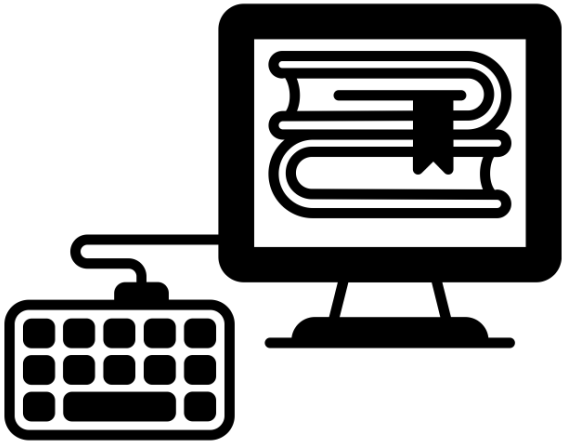
- Interactive lectures
- Discussions
- Small group work
- On demand demonstrations and guides
- Practical case studies using real datasets for surveillance and investigation



Informatics for FETP Materials

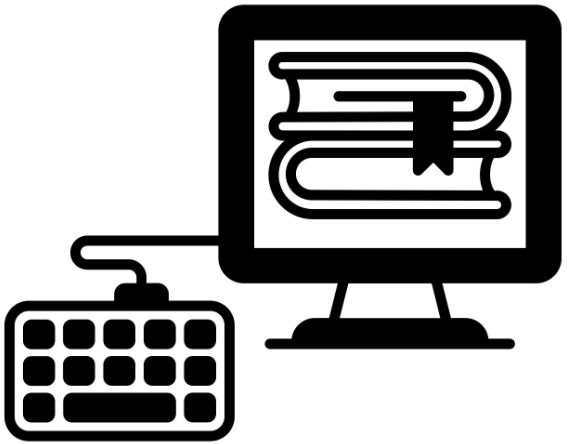
Available in hard and digital copy.

- Goals and objectives
- Schedule
- Unit learning objectives
- Presentation slides and notes
- Demonstration videos and guides
- Case study datasets and guide

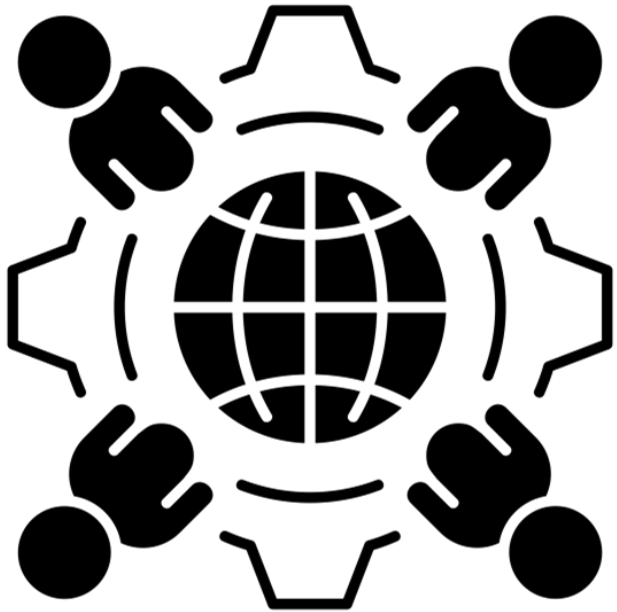


Accessing Digital Materials: Jupyter Notebook

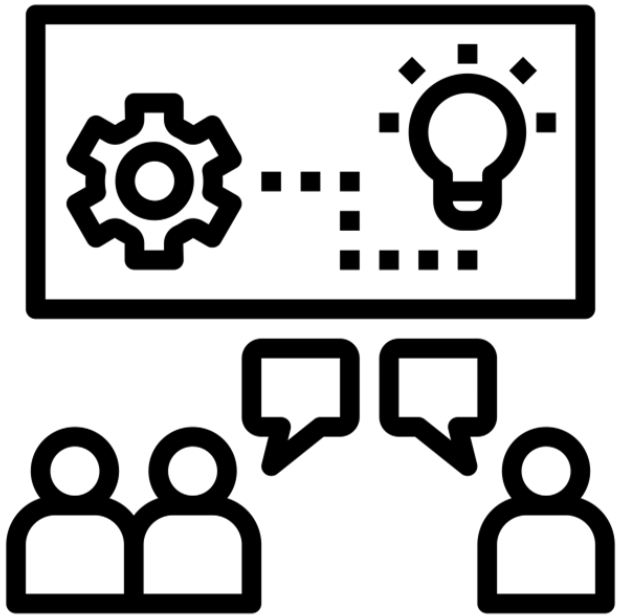
- What is a “Jupyter Notebook?”
- What materials are in the Informatics for FETP Jupyter Notebook?
- How to access and set up a Jupyter Notebook



Ground Rules/ Norms



Housekeeping



Unit 1:
Module Overview and
Introduction to Informatics for FETP

Unit 1 Goal and Learning Objectives

Goal: Relate key informatics principles, systems, and tools to the public health surveillance cycle and system

Learning Objectives:

- Describe the impact of the digital revolution on public health
- Describe how key informatics principles, systems, and tools can improve public health surveillance and outbreak investigations.
- Explain how the transformation of raw data to meaningful information, knowledge, and wisdom necessary supports data-driven action.
- Match your country’s health information system landscape and current trends to surveillance system components.

The Digital Revolution

- Accelerating since the mid-20th century.
  - Evolving at an exponential rate.
  - Impacting nearly every industry in every country.
- Transforming public health.
  - Data are the lifeblood of effective public health at all levels.
  - Data are being digitized and transmitted through ICTs.
  - Technology giants are leading innovations, with global IT spending exceeding $5 trillion.
- Presents both challenges and opportunities
- We must leverage digital technologies and harness their potential to improve patient and public health outcomes.

When well-planned, designed, implemented, managed, and used, digitization promises:

- Improve health care outcomes, access, and quality and patient safety
- Reduce health care costs, duplication, and waste
- Optimize resource allocation
- Empower individuals and communities to be informed consumers
- Improve the completeness, timeliness, and accuracy of data
- Train health professionals
- Hasten translation of research findings into practice and policy

Digitization is essential for every organization in public health, from the Ministry of Health (MoH) and its departments to National Public Health Institutes to regional and district health offices and hospitals, health centers, and health posts.

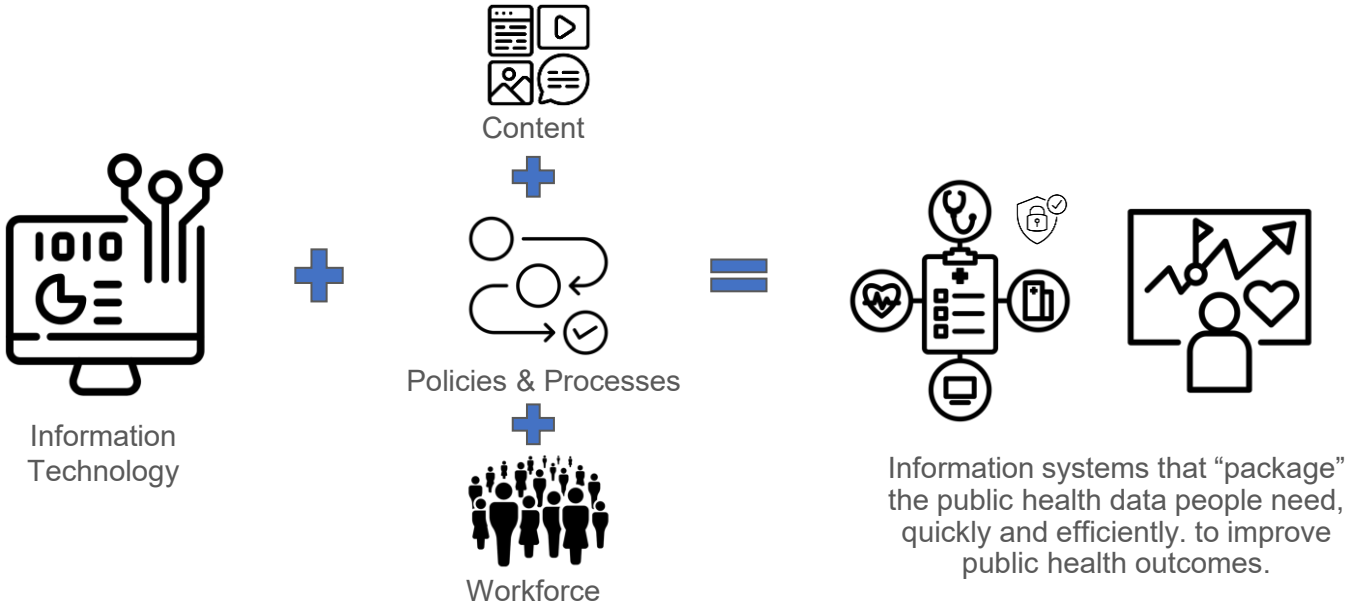
Why Informatics and Epidemiology?

- Field epidemiologists' ability to collect, analyze, interpret, and use data is critical to their everyday work.
- Each phase of the surveillance cycle, each step of an outbreak investigation has specific technology and information needs.
- Epidemiologists experience increased demands on their time and resource constraints at the same time.

Public health informatics has the potential to

- Decrease the time it takes from disease detection to reporting to response
- Allow real-time data for outbreak investigation and control

What is public health informatics?

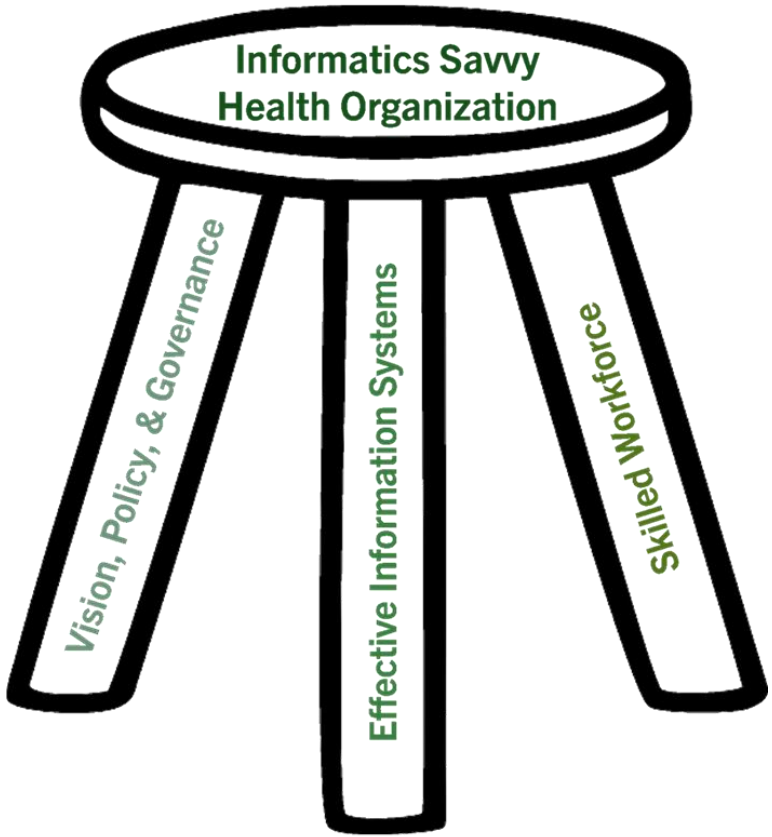


Public health informatics is the systematic application of information, computer science, and technology in areas of public health, (surveillance, prevention, preparedness, and health promotion) to capture, manage, analyze, disseminate, and use information to improve population-level health outcomes.[2]

Informatics Savvy Health Organization

Obtain, effectively use, and securely exchange information *electronically* to improve public health practice and population health outcomes

- Organization wide informatics vision, policy, and governance
- Skilled workforce
- Effective information systems



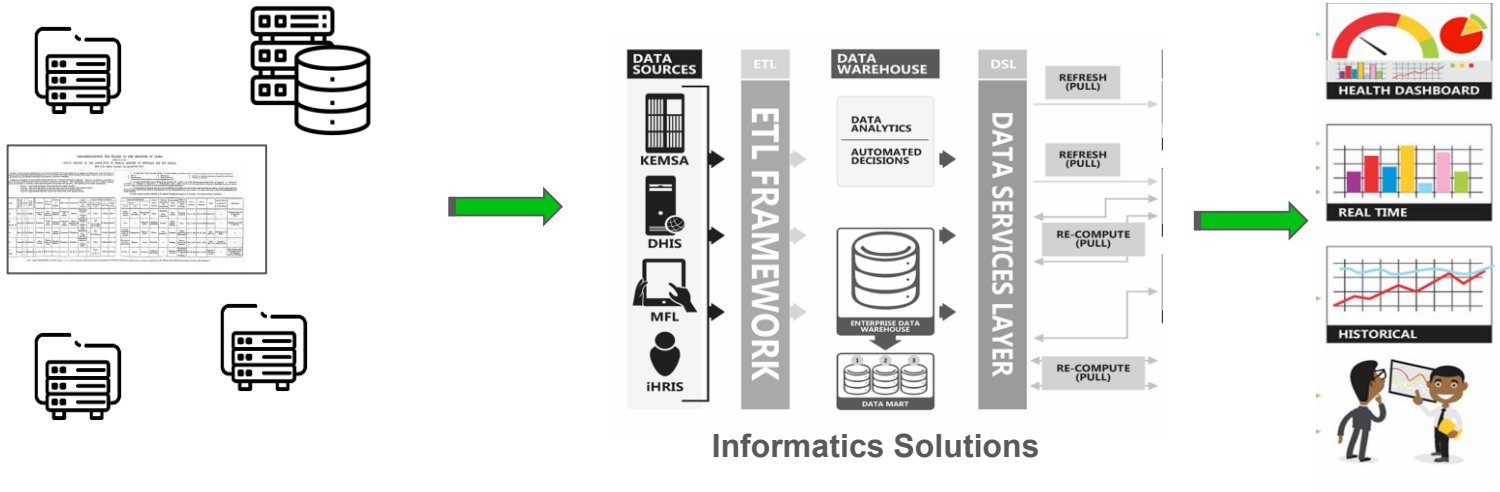
Science/data-driven (Informatics), people centric (Savvy), and organization focused (Health Organization)

What do informaticians do?

Informaticians work to strengthen public health organizations.

Architects plan buildings with the needs of a community in mind.

Informaticians plan, maintain, and improve data systems that produce the information public health professionals need to make sound public health decisions.





Laila's Problem: Data Overload

As a field epidemiologist, Laila is always on the lookout for potential outbreaks.

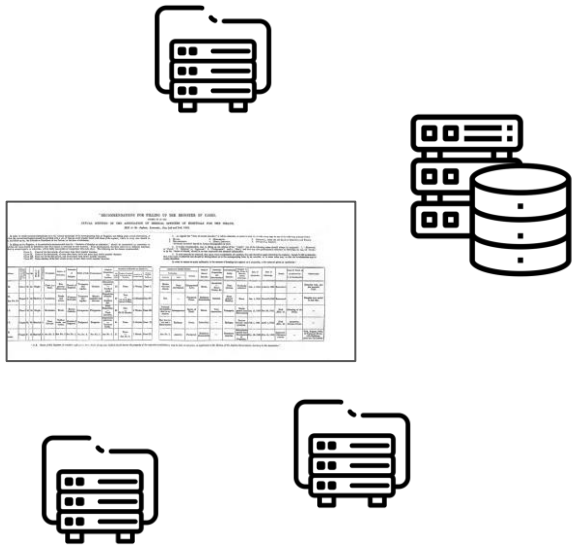
Her challenge?

Laila lives in the digital world. This means that she has a large volume of data coming to her at full speed from a wide variety of sources, some from paper forms or reports and others via electronic systems.

Locating, cleaning, and transforming the data she needs into information she can use can be time-consuming. She has many demands on her time.

How many of you can relate to this same challenge?

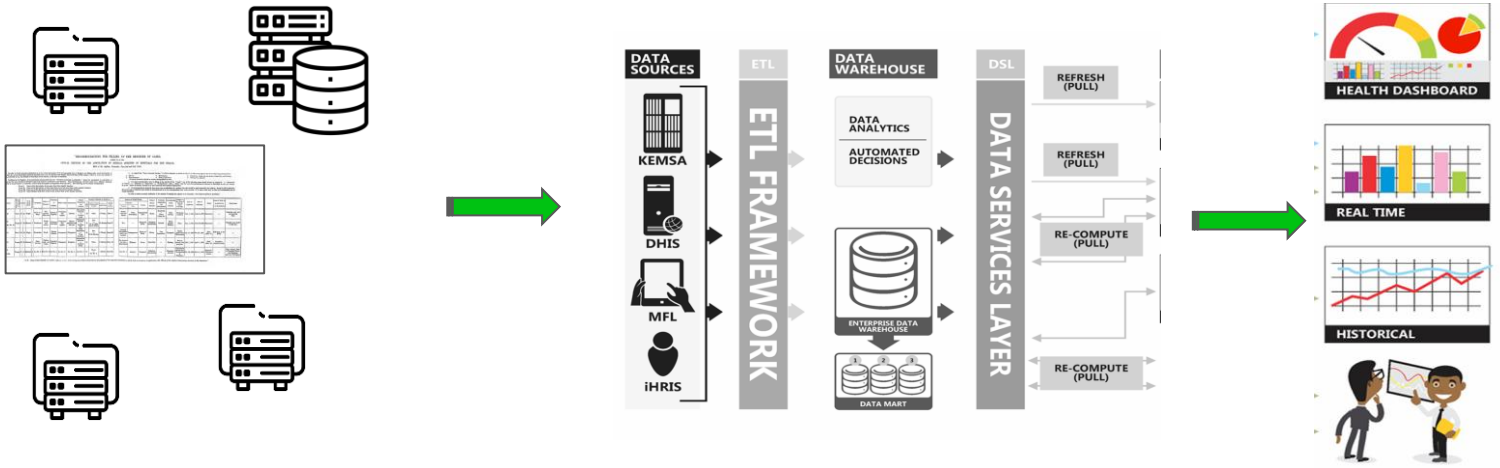
How are you solving this problem?



Current State



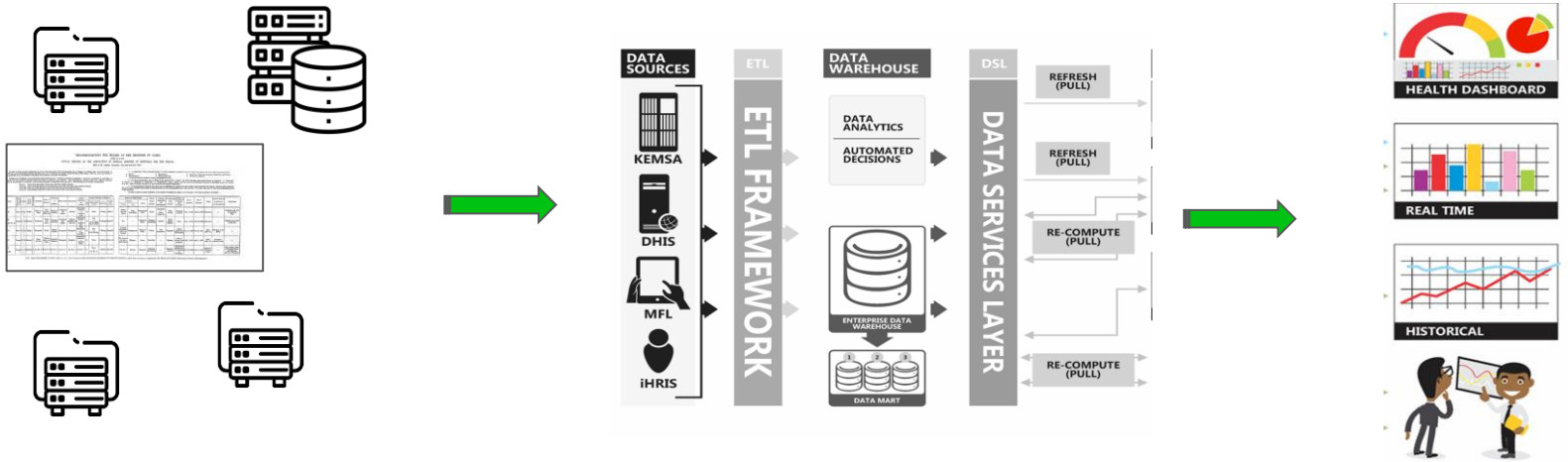
How can public health informatics solve Laila's problem?



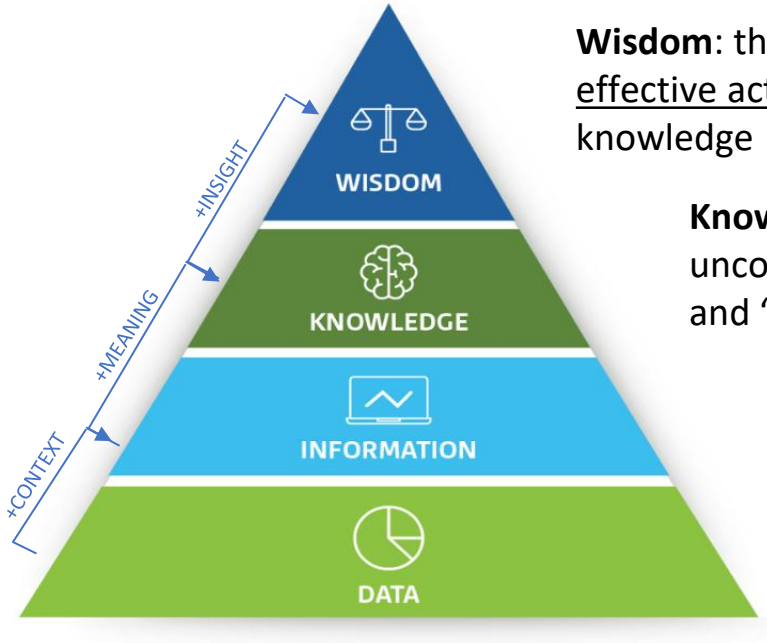
Informatics Solutions



How can public health informatics solve Laila's problem?



Translating and Transforming Data into Wisdom



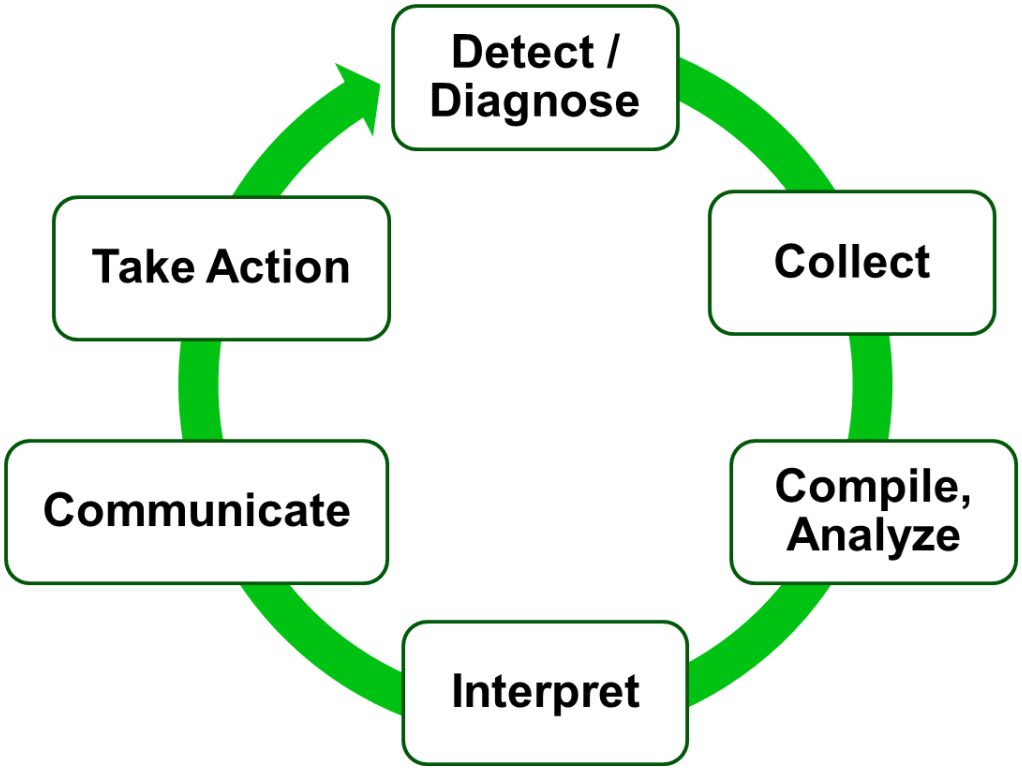
Wisdom: the ability to make well-informed decisions and take effective actions based on understanding of the underlying knowledge

Knowledge: analyzed and interpreted information to uncover patterns, trends & relationships. Uncovers "how" and "why."

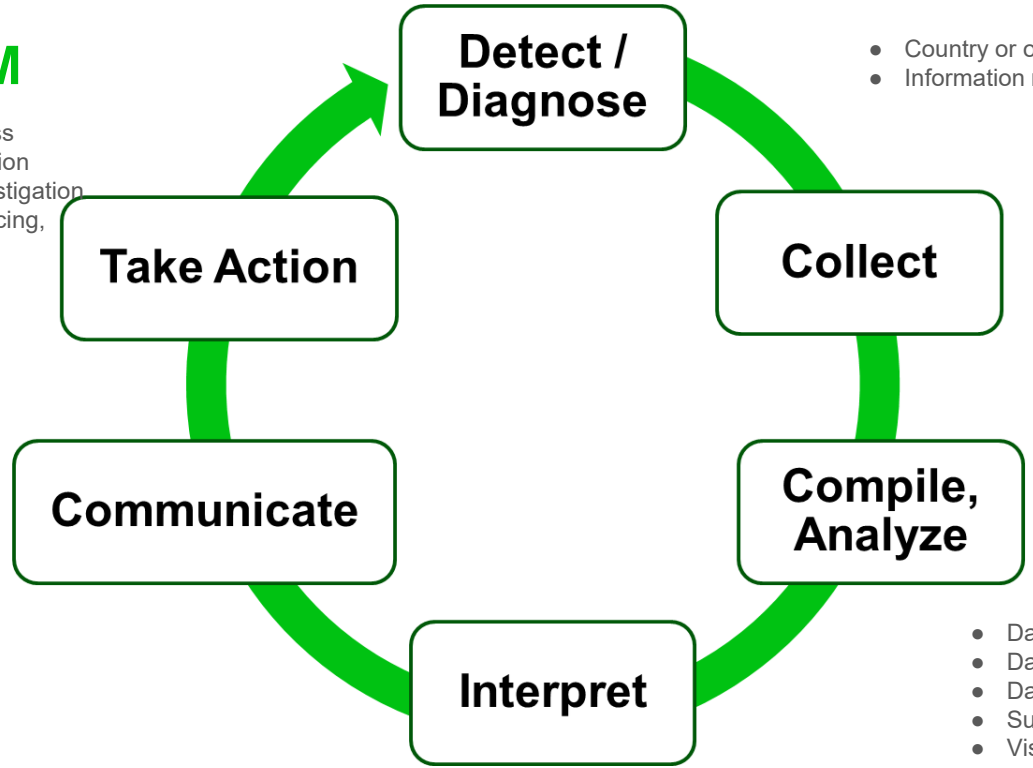
Information: organized, structure and contextualized data. Answers "person," "place," and "time."

Data: raw unprocessed facts and figures. Holds limited value in isolation.

Public Health Surveillance Cycle



Creating Value from Public Health Surveillance Data



WISDOM

- Impact, situation awareness
- Decision on short-term action
- Information needs for investigation
- Resource allocation (financing, workforce) for investigation

- Country or organizational context
- Information needs

- Data sources
- Data standards (variables, types, formats)
- Data quality
- Data security, privacy, & confidentiality

DATA

- Disseminate summary of surveillance data
- Alerts about outbreaks or events

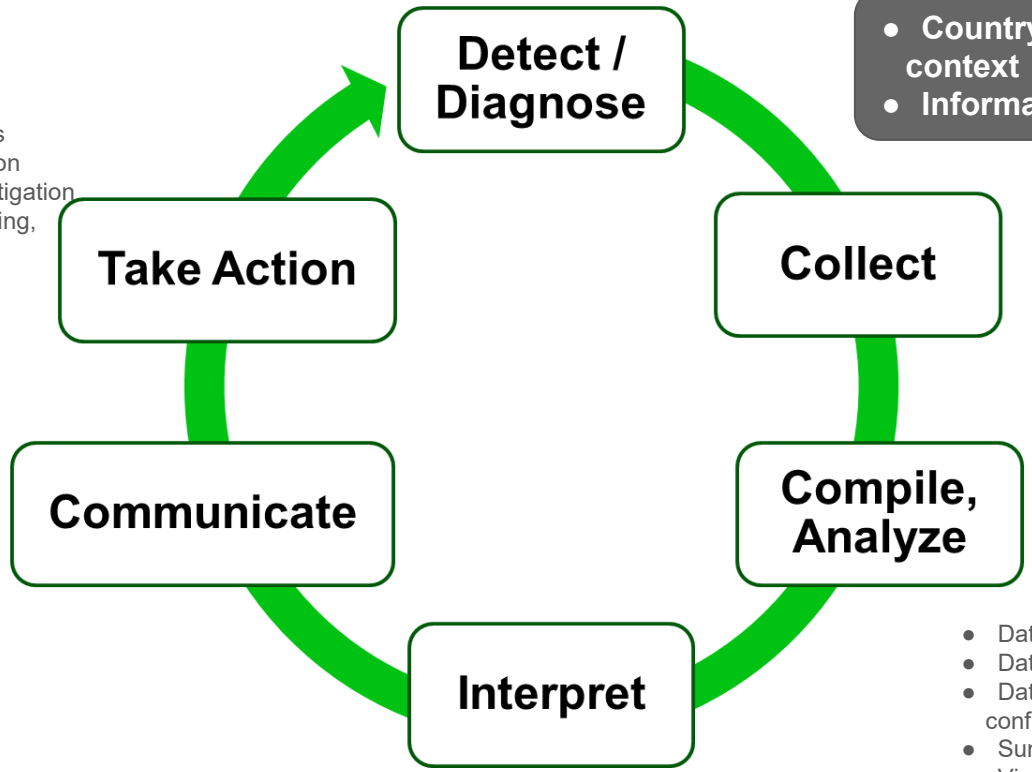
KNOWLEDGE

- Data organization & storage
- Data cleaning & transformation
- Data merging & exchange
- Summarize data
- Visualize data

- Describe and interpret data

INFORMATION

Creating Value from Public Health Surveillance Data



- **Country or organizational context**
- **Information needs**

- Data sources
- Data standards (variables, types, formats)
- Data quality
- Data cleaning & transformation

- Data organization & storage
- Data merging & exchange
- Data security, privacy, & confidentiality
- Summarize data
- Visualize data

- Describe and interpret data

- Disseminate summary of surveillance data
- Alerts about outbreaks or events

- Impact, situation awareness
- Decision on short-term action
- Information needs for investigation
- Resource allocation (financing, workforce) for investigation

Summary

- Informatics applies relevant information technology to content and processes that help epidemiologists use data to rapidly detect, diagnose, and take action to address public health emergencies.
- Informaticians specialize in translating data into information, which allows epidemiologists to apply their knowledge and turn into wisdom to solve epidemiological problems.
- By applying informatics to the public health surveillance cycle, epidemiologists can extract greater value from the data available to them.
- Applying informatics principles enables epidemiologists to use informatics systems and tools more effectively.

Taking the Pulse

What are three things you learned in this unit?



What questions do you still have?

